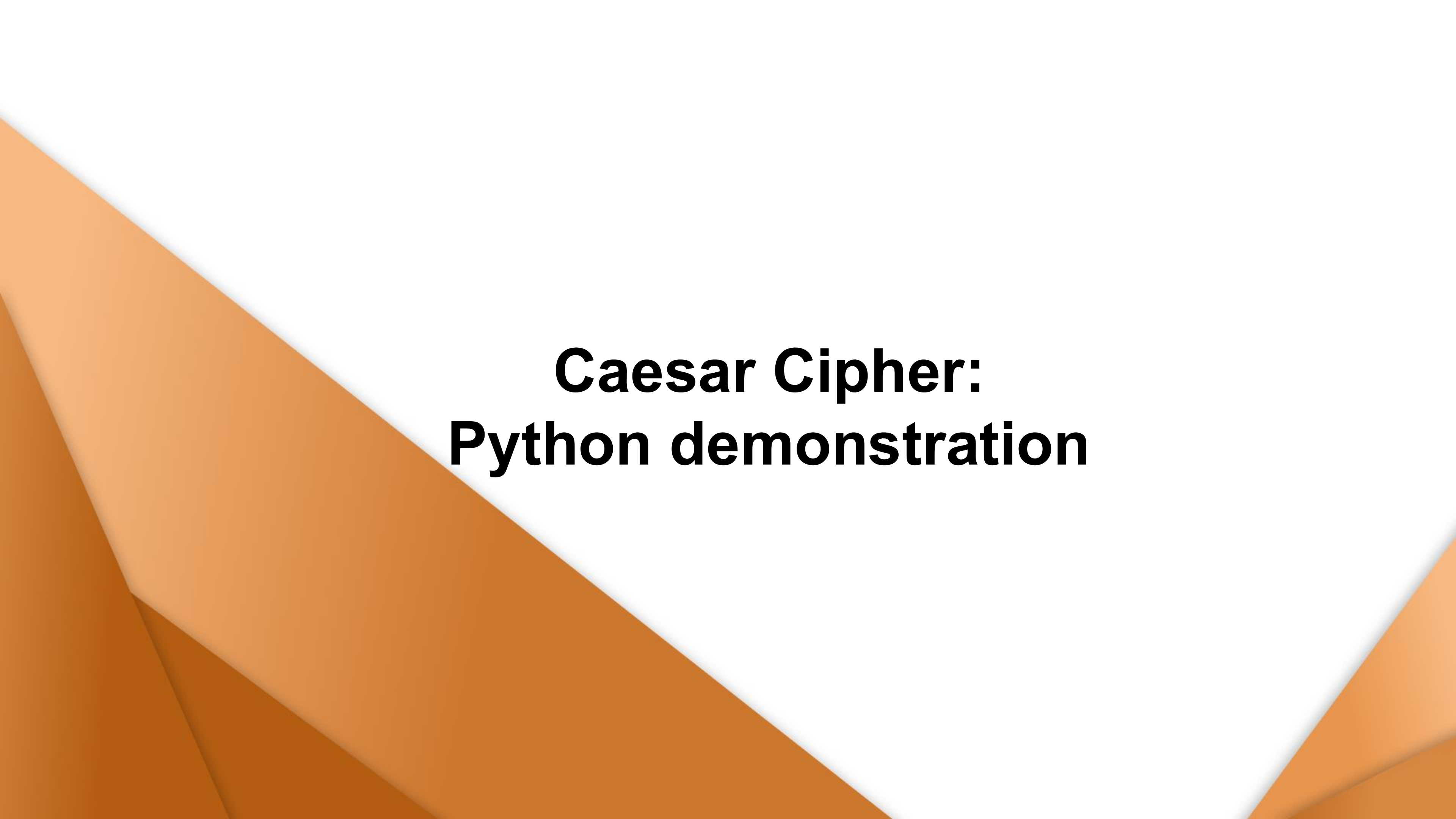




Caesar Cipher: Python demonstration

Learning objectives

- **Describe how the Caesar Cipher works**
- **Build encryption and decryption in Python**
- **Explain basic character shifting**

What is Caesar Cipher?

- **Substitution cipher that shifts letters**
- **Each letter is moved by a fixed number of places down the alphabet**
- **Named after Julius Caesar**

Encryption function

- **Shifts characters forward by 'n' positions**
- **Uses American Standard Code for Information Interchange (ASCII) to transform characters**
- **Leaves spaces and punctuation unchanged**

Here is the code:

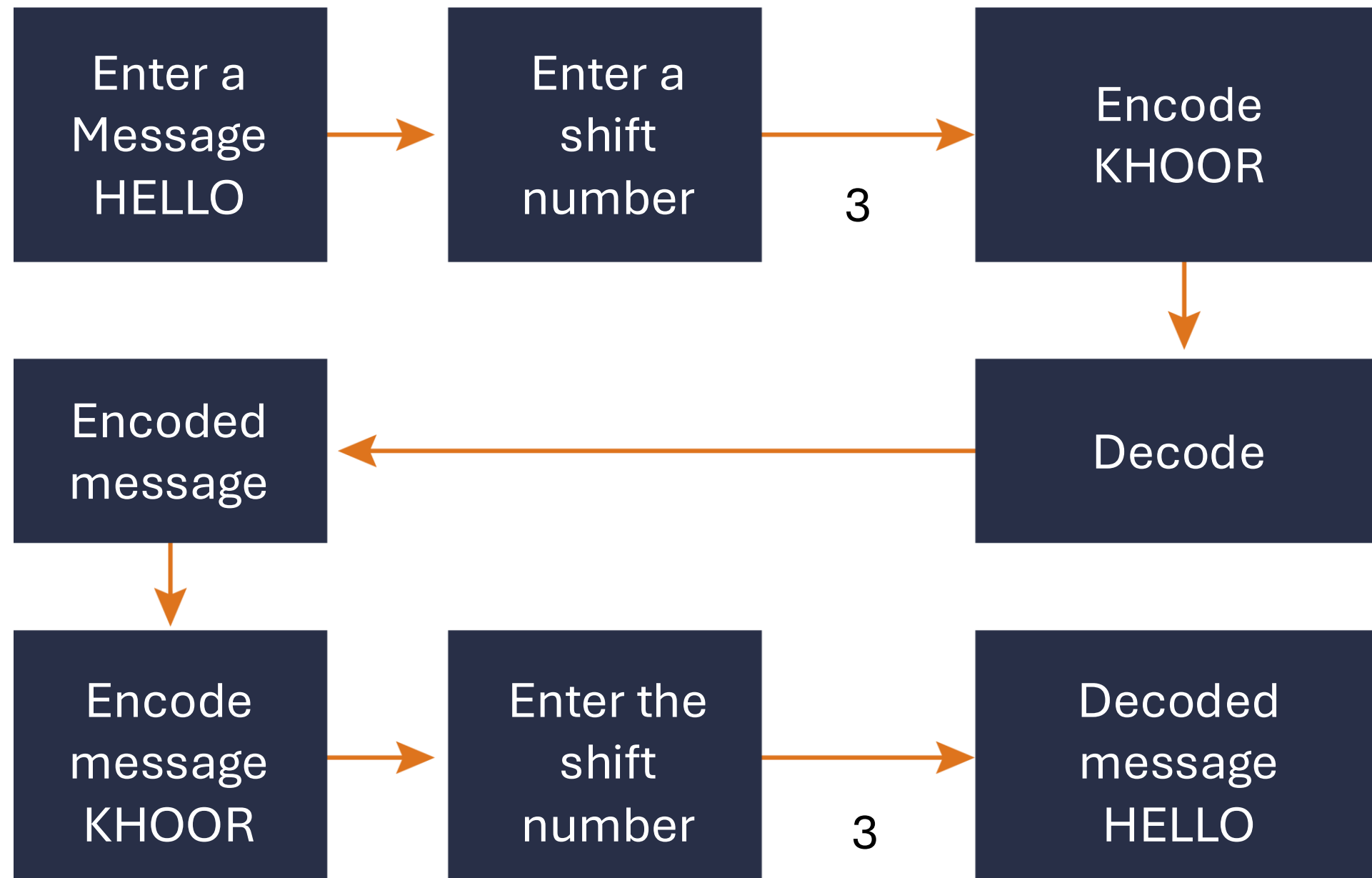
```
def encrypt(text, shift):  
    result = ""  
    for char in text:  
        if char.isalpha():  
            base = ord('A') if char.isupper() else ord('a')  
            result += chr((ord(char) - base + shift) % 26 +  
base)  
        else:  
            result += char  
    return result
```

Decryption function

- **Shifts characters backwards**
- **Same logic as encryption with a negative shift**
- **Reuses encryption function for simplicity**

```
def decrypt(text, shift):  
    return encrypt(text, -shift)
```

Running the program



```
message = "Hello, world!"  
shift = 3
```

```
encrypted = encrypt(message, shift)  
decrypted = decrypt(encrypted, shift)
```

```
print("Original:", message)  
print("Encrypted:", encrypted)  
print("Decrypted:", decrypted)
```

Summary

- **Built a Caesar Cipher in Python**
- **Learned basic character handling and logic**

Next

- **More advanced cryptographic techniques to follow**